

FluxCast

FORECASTING THE FLOW
OF RENEWABLE POWER

TURNING DATA INTO
A BRIGHTER TOMORROW.



SOLAR



WIND



FORECAST



A CLEANER
TOMORROW

Team #E4

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Renewable energy is clean – but **unpredictable**.

That unpredictability costs grids money, wastes clean power, and increases reliance on fossil backup.



Small weather changes cascade into real operational and financial costs.



Wasted clean energy

Over-generation gets curtailed because operators can't act on it in time.



Costly last-minute backup

Under-forecast periods force expensive, high-emission fossil backup generation.

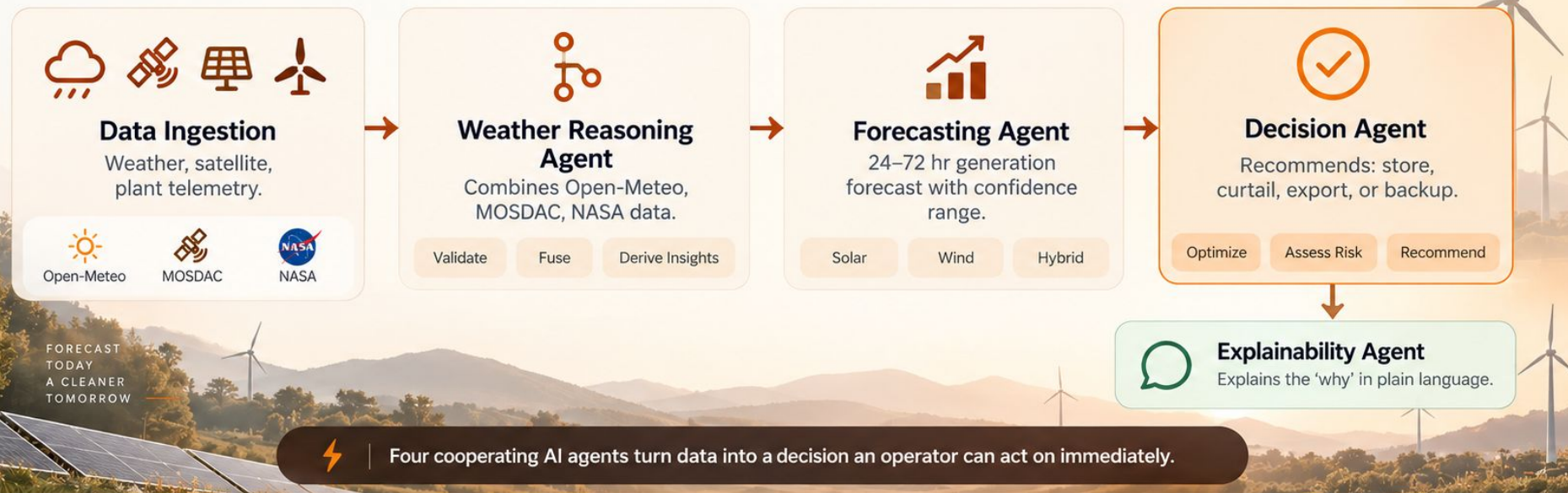


No actionable guidance

Existing tools predict a number, not a decision, leaving operators to guess what to actually do.

From raw weather data to a clear **grid decision**.

Fluxcast doesn't just forecast — it recommends what to do next.



Higher renewable utilization



Lower operating costs



Reduced emissions



More stable and reliable grids

What better forecasting is worth.

More clean energy on the grid. Lower costs. A more reliable tomorrow.



↓ 30%

Curtailment

Less clean energy wasted.



↓ 25%

Fossil backup usage

Lower emissions and operating cost.



+20%

Forecast accuracy improvement

vs. naive baseline
(24-72 hr horizon).



24-72 hr

Forecast horizon

Ahead of time for
proactive grid decisions.

Illustrative targets based on published industry curtailment/forecast-error benchmarks; to be validated with pilot data.



Turning better forecasts into a **cleaner, more reliable** and **more cost-efficient** grid.

FORECAST
TODAY
A CLEANER
TOMORROW

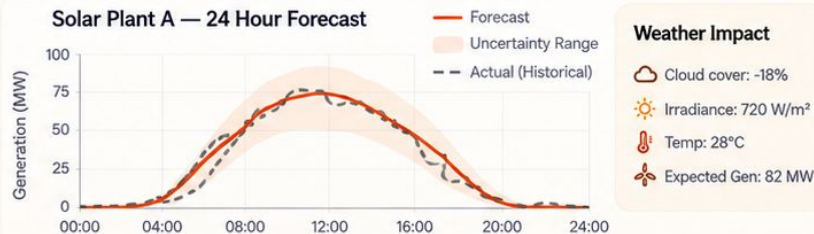
Not just a forecast — a forecasting partner.

Two key differentiators that turn predictions into real-world impact.



Plant-Aware Forecasting

Predicts plant-level and portfolio-level power generation, and explains exactly how weather conditions at each site affect grid supply — not a generic regional estimate.



Site-specific insights for more accurate and actionable forecasts.



Proactive, Not Reactive AI

Fluxcast's agents don't wait to be asked — they continuously predict upcoming generation and automatically alert operators to urgent risk windows before they become costly problems.



High Deficit Risk Detected

2 hours ago

Solar Plant A is expected to generate 40% lower power tomorrow (12:00–17:00) due to high cloud cover.



Expected Deficit
48 MW



Time Window
12:00 – 17:00



Recommended Action
Dispatch storage



Early alerts. Clear actions. Lower costs.

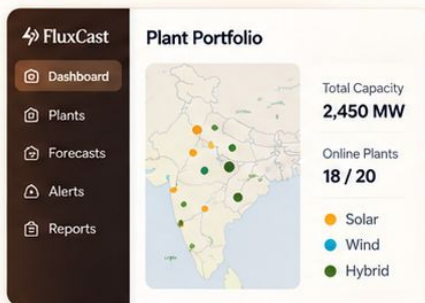


Fluxcast acts like an **analyst on your team** — not just a dashboard you have to keep checking.

A day in the life of a grid operator using Fluxcast.

From visibility to action — in a few clicks.

1 Log in



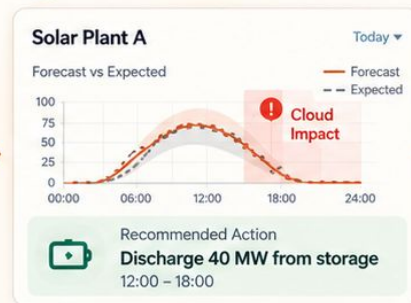
Operator opens the portfolio dashboard, sees all plants on the map.

2 Spot a risk



An alert flags an incoming cloud system affecting a specific solar site.

3 Check the recommendation



Fluxcast suggests discharging stored battery power ahead of the dip.

4 Act with confidence



Operator approves the action directly from the dashboard, avoiding a shortfall.

PEOPLE
POWER
POSSIBILITIES



From raw risk to resolved action in four steps — **no manual digging through weather reports.**

Duo Subscription Model

A proven SaaS model for today, and a high-value API layer for tomorrow.



SaaS Subscription

For utilities and grid operators

Utilities and grid operators pay a monthly/annual fee based on the number of plants or total MW capacity monitored — predictable recurring revenue that scales with customer size.

- ✓ Portfolio dashboard (multi-plant view)
- ✓ 24–72 hr generation forecasts
- ✓ Risk alerts and actionable recommendations
- ✓ Historical analysis and reports
- ✓ Role-based access for teams

Target
customers



Utilities & Grid Operators
(GENCOs, DISCOMs, IPPs)



Premium API Access

For integration and high-value use cases

Direct API access to Fluxcast's forecasting and recommendation engine, for integration into a utility's existing SCADA/EMS software or a trading desk's own tools.

- ✓ Forecasting + recommendation API
- ✓ Easy integration with SCADA/EMS systems
- ✓ Real-time and batch data access
- ✓ Custom endpoints for trading and analytics
- ✓ Enterprise support and SLAs



Moat: The API doesn't just return a weather number — it returns the output of our full agentic workflow. A competitor could copy a weather API call, but not our trained multi-agent decision layer built on proprietary plant data.

Target
customers



Energy Trading Desks
Market signals & strategy



EMS / SCADA Vendors
Embed Fluxcast into
existing platforms

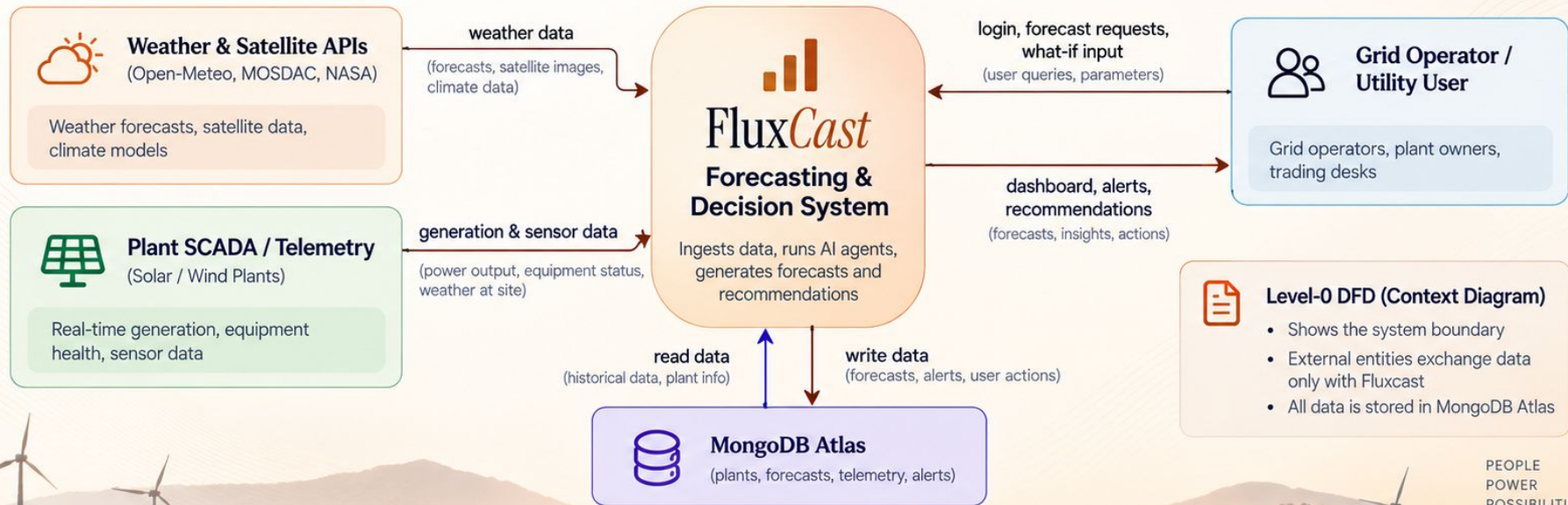


Recurring SaaS revenue today. **High-margin, defensible API revenue tomorrow.**

PEOPLE
POWER
POSSIBILITIES

High level architecture.

A single context-level view: external entities, the Fluxcast process, and the data stores it relies on.



PEOPLE
POWER
POSSIBILITIES



One governed process. Multiple data sources. Smarter decisions for a **cleaner, more reliable grid.**

FluxCast Feasibility

Built with today's technology. Designed for real-world adoption.



Technical Feasibility

Mature tech. No exotic infrastructure.

Backend		Node.js + Express
Frontend		React
Database		MongoDB Atlas
AI Agent Framework		LangChain / LangGraph
Data Sources	  	Open-Meteo MOSDAC NASA

Built entirely on widely-used technology and free/public data sources — no proprietary hardware required.

Feasibility



High



Financial Feasibility

Low upfront cost. Scalable revenue.



Cloud-hosted

No large infrastructure investment required.



Usage-based APIs

Leverage cost-effective, public data sources.



Scalable subscription model

Revenue scales with number of plants or total MW capacity.

Low upfront cost and a clear path to recurring revenue that grows with customer size.

Feasibility



High



Operational Feasibility

Easy to adopt. Minimal disruption.



Works with existing systems

Designed to run alongside current SCADA/EMS infrastructure.



Start small

Pilot on a subset of plants before wider rollout.



Minimal disruption

No changes to existing operations required.

Fits into existing workflows, with a low-risk path to adoption.

Feasibility



Medium



Technically achievable. Financially viable. Operationally practical. **A clear path from prototype to production.**